SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Dhivya Dharshini S *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dhivya

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2.25	2.25	2.25	2.25
Explanation	25%	2.25	2.25	2.25	2.25
Application	20%	1.80	1.80	1.80	1.80
Organization	15%	1.35	1.35	1.35	1.35
Accuracy	10%	0.9	0.9	0.9	0.9
Clarity	5%	0.45	0.45	0.45	0.45

Total Marks 36 **/40**

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

SECTION / DEPARTMENT

CSA0710 - Computer Networks

COI ATI - SHORT ANSWER

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36/100

6) Given:

Original file size = 900 MB

compression = 40%.

Encryption overhead = 5%.

1. **calculate compressed file size:**

compression reduces the file size by 40%.

$$\text{compressed size} = \text{original size} \times \frac{(100-40)}{100}$$

$$= 900 \times \frac{60}{100}$$

compressed file size.	= 540 MB

2. **calculate encrypted file size:**

Encryptions adds 5% overhead to the compressed size.

$$\text{Encrypted size} = 540 + \left[540 \times \frac{5}{100}\right]$$

encrypted file size	= 567 MB

3. calculate total percentage reduction

Reduction = original - encrypted size

= 900 - 567

= 333 MB

Percentage reduction = (Reduction / original size)

$$= \frac{333}{900} \times 100$$

total percentage reduction	= 37%

The presentation layer improves transmission efficiency by compressing data and reducing the amount of data transferred over the network.

It also supports encryption to protect sensitive information from unauthorised data.

Therefore, it helps improve bandwidth utilisation and data security.

Given,

Total data = 3 GiB

Throughput = 120 Mbps

Request size = 3MB

∵ 1GB = 1024 MB

1. calculate the total no. of. requests

Total data = 3 x 1024

= 3072 MB

No. of. requests = Total data / request size

= 3072 / 3

= 1024 requests

2. calculate transmission time

converting the total data into megabites

1MB = 8 megabites

Total data = 3072 x 8

= 24,576 megabits.

Transmission time = Total data / Throughput

= 24,576 / 120 = 204.8 seconds

How application layer provides reliable services to users

The application layer provides network services to user application through protocol such as FTP.

FTP supports file transfer between computers. It allows user to upload and download files, where errors handling and reliable delivery depend on the underlying transport protocol, typically TCP.

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8) How end-to-end delay experienced by packets:

Usually end to end delay is the total time taken by a packet to travel from a sender to the receiver through network.

When data travels across a network, it experiences different type of delays. It includes processing delay at different OSI layers, transmission delay and propagation delay.

The total end-to-end delay is calculated by adding all the given delays.

Formula:

Total delay = [processing + transmission + propagation] delay

Processing delay is the sum of the delay given for all OSI layers

Given datas:

application layer = 4ms

Transport layer = 3ms

Network layer = 5ms

Data link layer = 2ms

Physical layer = 1ms

Propagation layer = 16ms

Transmission layer = 12ms

calculation:

1. Calculating the total processing delay:

$D_t = 4 + 3 + 5 + 2 + 1$

$= 15$ ms

Total processing delay = 15ms

2. Identify the other delays

Transmission delay = 12 ms

Propagation delay = 18 ms

3. End-to end delays applying:

$Delay_{total} = 15 + 12 + 18$

$= 45$ ms.

Total delay = 45 milliseconds.

OSI layer contribution to communication:

Application layer

Transport layer

Network layer

Data link layer

Physical layer.

conclusion :

The packet experiences a total processing delay of 15ms across the five OSI layers, a transmission delay of 12 ms, propagation delay of 18ms, Adding these, we get the value of 45 ms.

9) Given :

Useful data = 80MB

Frame size = 1600 bytes

protocol overhead per frame = 80 bytes.

Assuming the 1600 byte frame size includes 80 - byte overhead.

1. calculate useful data per frame

Useful data frame = Frame size - overhead.

= 1600 - 80

= 1520 bytes

2. calculate the total number of frames.

Assume 1MB = 1000,000 bytes

Total useful data = 80 × 10,00,000

= 80,000,000 bytes

Number of frames = $\frac{\text{Total useful data}}{\text{useful data per frame.}}$

= $\frac{80,000000}{1522}$

Total frames = 52,631.58

since, a complete frame is required for the remaining data, round up.

3. calculate total overhead transmitted

Total overhead = No. of. frames × overhead per frame

= 52632 × 80

= 4,210,560 bytes

4. Calculate transmission delay:

Transmission

$$\text{Efficiency} = \frac{\text{Useful data}}{\text{Total transmitted data}} \times 100$$

Total transmitted data = useful data + total overhead

$= 80,000,000 + 4,210,560$

$= 84,210,560$ bytes

$$\text{Efficiency} = \frac{80,000,000}{84,210,560} \times 100$$

$\simeq 95\%$.

Protocol overhead consists of additional information such as headers and trailers, required for addressing, routing, error detection and reliable communications.